

BLUESTOCK MUTUAL FUND ANALYTICS

Final Project Report

End-to-end ETL, performance analytics, investor intelligence, portfolio concentration and Power BI dashboard

Prepared for the Bluestock MF Capstone

Analytical universe: 34 schemes with usable NAV observations

Release: v1.0

1. Executive Summary

This project delivers a complete mutual-fund analytics workflow from raw CSV ingestion through live NAV integration, validation, SQLite storage, statistical analysis and an interactive Power BI dashboard. The final analytical universe contains 34 schemes with usable NAV history, supported by 35 metadata records, 32,778 investor transactions and 322 portfolio holding records.

Performance analysis shows a strong separation between absolute return leaders and risk-adjusted leaders. The top three 3-year return schemes in the supplied performance dataset are SBI Small Cap Fund – Regular Growth (23.39%), SBI Small Cap Fund – Direct Growth (23.14%) and ABSL Small Cap Fund – Regular Growth (22.38%). Meanwhile, the highest Sharpe ratios are concentrated in low-risk/liquid and short-duration debt schemes, demonstrating why return and risk-adjusted return should be viewed together.

The advanced analytics layer adds historical VaR/CVaR, rolling 90-day Sharpe, investor cohort analysis, SIP continuity monitoring and sector HHI concentration. The SIP continuity analysis identifies 38 investors with six or more SIP transactions; all 38 are classified as at-risk under the >35-day average-gap rule in the supplied output. This is a small but actionable retention cohort.

The Power BI model contains four pages—Executive Overview, Fund Performance, Investor Analysis, and Portfolio & Market—providing an interactive presentation layer for the analytical outputs.

Overall, the project demonstrates a full analytics lifecycle: reliable data engineering, reproducible calculations, business-oriented risk metrics and an executive dashboard.

2. Problem Statement, Objectives and Scope

The objective was to convert heterogeneous mutual-fund datasets into a reusable analytics product capable of answering three questions: how the funds have performed, how much risk investors are taking, and what portfolio/investor behaviours deserve attention.

Core objectives: ingest and validate source data; integrate live NAV data; build EDA; calculate CAGR, Sharpe, Sortino, Alpha, Beta and drawdown; produce VaR/CVaR and rolling Sharpe; analyse investor cohorts and SIP continuity; measure sector concentration with HHI; provide risk-based recommendations; and communicate the results through Power BI and final documentation.

Scope: the supplied analytical NAV universe contains 34 usable schemes. Metadata contains one additional record without usable NAV observations, which is retained for reference but excluded from NAV-return calculations. Benchmark analysis uses the supplied benchmark-index dataset and the NIFTY100 series where overlapping observations are available.

Key methodological conventions: daily returns are computed as $\text{NAV}(t)/\text{NAV}(t-1)-1$; Sharpe uses a 6.5% annual risk-free proxy and 252 trading days; Alpha is annualized from the regression intercept; Historical VaR is the 5th percentile and CVaR is the mean of returns below that threshold; HHI is the sum of squared sector weights.

3. Data Sources and Data Inventory

The project uses ten canonical datasets plus live NAV data. The ETL logs confirm successful ingestion and row-level validation for the canonical files.

Dataset	Rows	Columns	Purpose
Fund metadata	35	7	Scheme identity, fund house, category and ISINs
NAV history	97,828*	3	Historical daily NAV observations
AUM by fund house	90	5	Fund-house AUM trend
Monthly SIP inflows	48	6	Industry SIP trend
Category inflows	144	3	Category-level inflow comparison
Folio count	21	6	Investor participation trend
Scheme performance	40	19	Return/risk reference metrics
Investor transactions	32,778	13	Investor behaviour and transaction amounts
Portfolio holdings	322	8	Security and sector allocation
Benchmark indices	8,050 raw / 3,171 processed	13	Market benchmark series

* The notebook analytics snapshot uses the long NAV-history file; the master ETL canonical processed file is a shorter filtered snapshot. This distinction is documented to preserve reproducibility and avoid silently mixing snapshots.

4. ETL Design and Data Quality

The master pipeline executes five stages in order: Data Ingestion → Live NAV Fetch → Data Preprocessing → Data Validation → SQLite Loading. Each stage is implemented as a separate Python module and invoked by `run_pipeline.py` using project-relative pathlib paths.

Observed successful run: all ingestion datasets were loaded; live NAV was updated for 34 schemes; preprocessing completed; validation passed; and SQLite loading verified every table. The final ETL log ends with “ALL PIPELINE STAGES COMPLETED SUCCESSFULLY”.

Validation point	Result
Metadata rows after cleaning	35
NAV schemes with usable observations	34
Investor transactions	32,778
Portfolio holdings	322
Benchmark processed rows	3,171
SQLite loading	Verified for all canonical tables
Metadata-only extra scheme	Excluded from NAV analytics

The main remaining data-quality caveat is that the project contains multiple snapshots of the NAV history. The final report treats the notebook analytical snapshot and the ETL canonical snapshot as distinct rather than claiming they are identical.

5. Exploratory Data Analysis — Market and Fund Landscape

EDA established the shape, completeness and distribution of the available fund and market data before performance modelling. The analysis identified 35 metadata rows, 34 NAV-active schemes and 40 reference scheme-performance rows.

SBI Mutual Fund is highlighted as one of the strongest industry players by AUM in the supplied AUM series, with reported AUM increasing from ₹630,000 crore in 2022 to ₹1,250,000 crore in 2025. The AUM analysis provides useful industry context for scheme-level performance.

Portfolio holdings show a broad but financially concentrated sector footprint. Banking is the largest aggregated sector allocation in the supplied holdings data, followed by IT and Pharma.

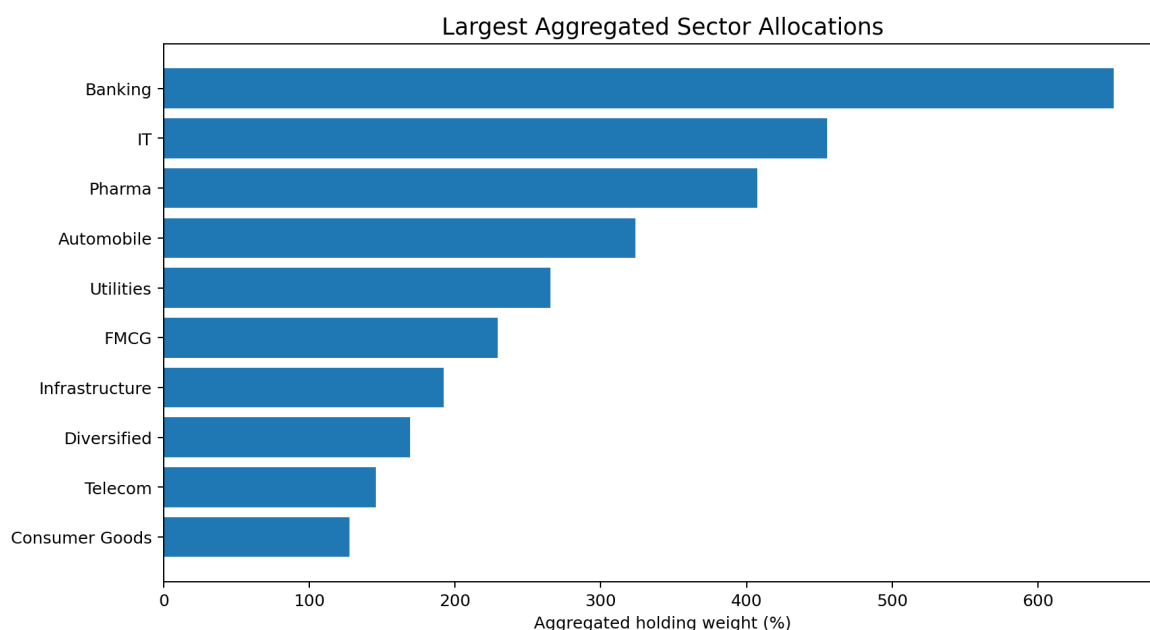


Figure 1. Largest aggregated sector allocations across the portfolio holdings dataset.

6. Exploratory Data Analysis — Investor Behaviour

The investor dataset contains 32,778 transactions across 5,000 unique investors. SIP transactions account for 19,716 records across 4,762 unique SIP investors.

The cohort analysis groups investors by the year of their first recorded transaction. The 2024 cohort contains 3,976 investors in the notebook cohort summary and accounts for approximately █1.349 billion in the cohort-specific invested amount reported there. The 2025 cohort contains 416 investors and approximately █52.5 million in the corresponding summary.

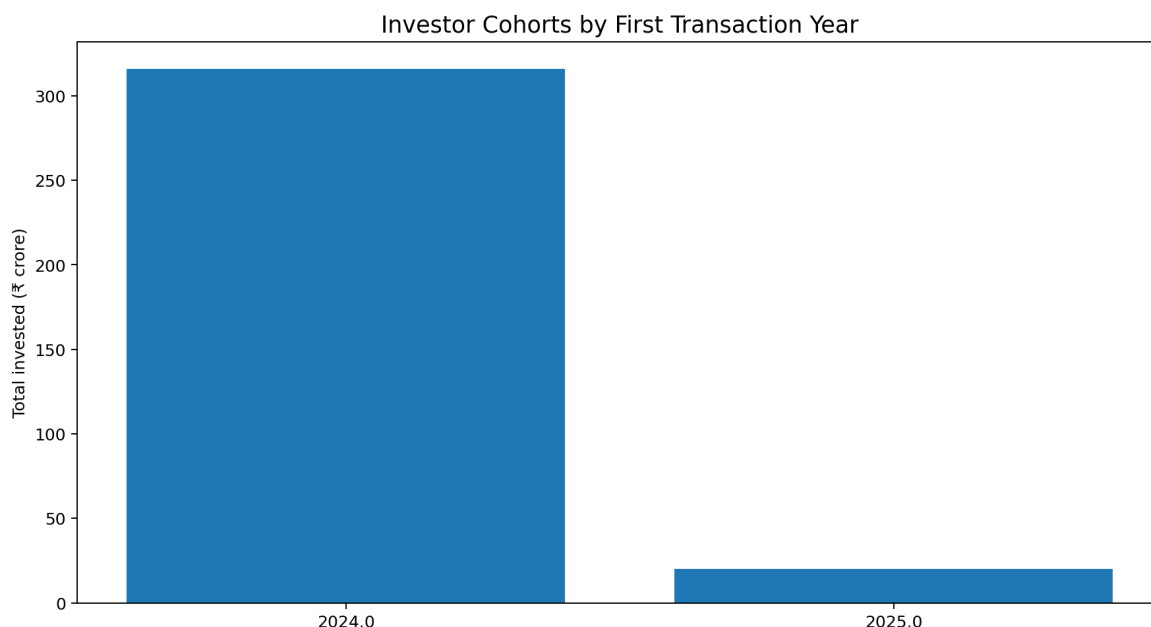


Figure 2. Cohort investment scale using the transaction-level cohort calculation.

The EDA also indicates a meaningful geographic and demographic dimension: the supplied notebook notes that T30 cities contribute a significant share of investment while B30 participation demonstrates broader geographic reach.

7. Performance Analysis

The performance layer combines absolute returns with risk-adjusted metrics. The supplied scheme-performance dataset contains 40 scheme rows and provides 1-year, 3-year and 5-year returns, Sharpe, Sortino, volatility, maximum drawdown, expense ratio and risk grade.

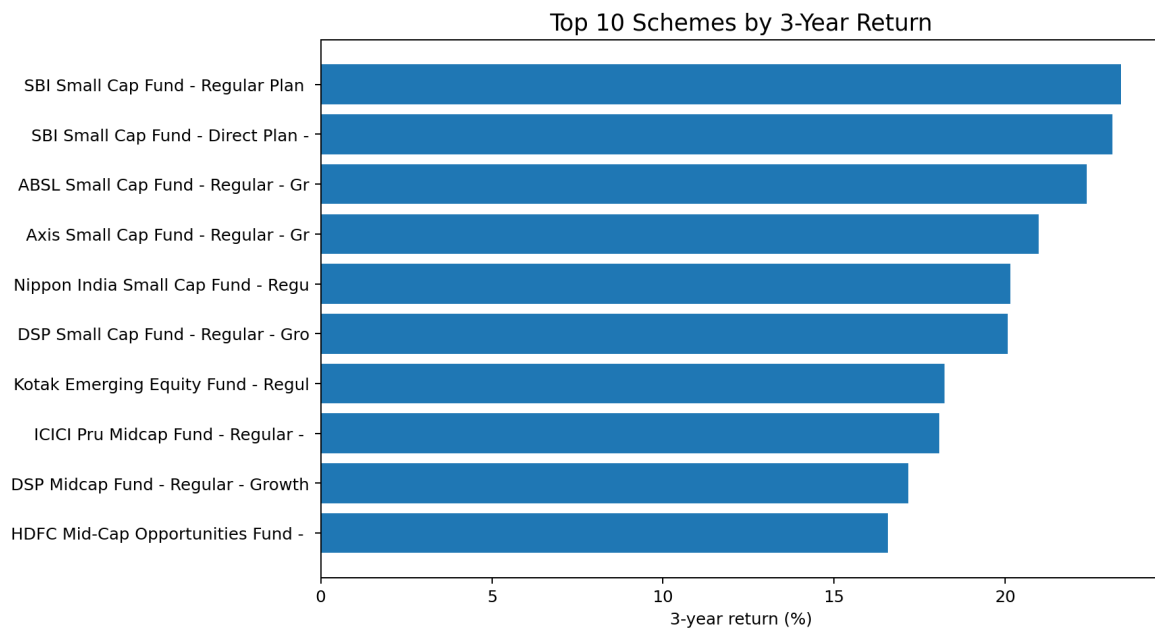


Figure 3. Top 10 schemes by 3-year return.

Rank	Scheme	3Y return	Risk grade
1	SBI Small Cap Fund - Regular Plan - Growth	23.39%	Very High
2	SBI Small Cap Fund - Direct Plan - Growth	23.14%	Very High
3	ABSL Small Cap Fund - Regular - Growth	22.38%	Very High
4	Axis Small Cap Fund - Regular - Growth	20.98%	Very High
5	Nippon India Small Cap Fund - Regular - Growth	20.15%	Very High

Absolute-return leadership is dominated by small-cap exposure in the supplied reference metrics. This is consistent with the higher risk grades attached to the leading return schemes and reinforces the need for risk-adjusted comparisons.

8. Risk-Adjusted Performance — Sharpe and Sortino

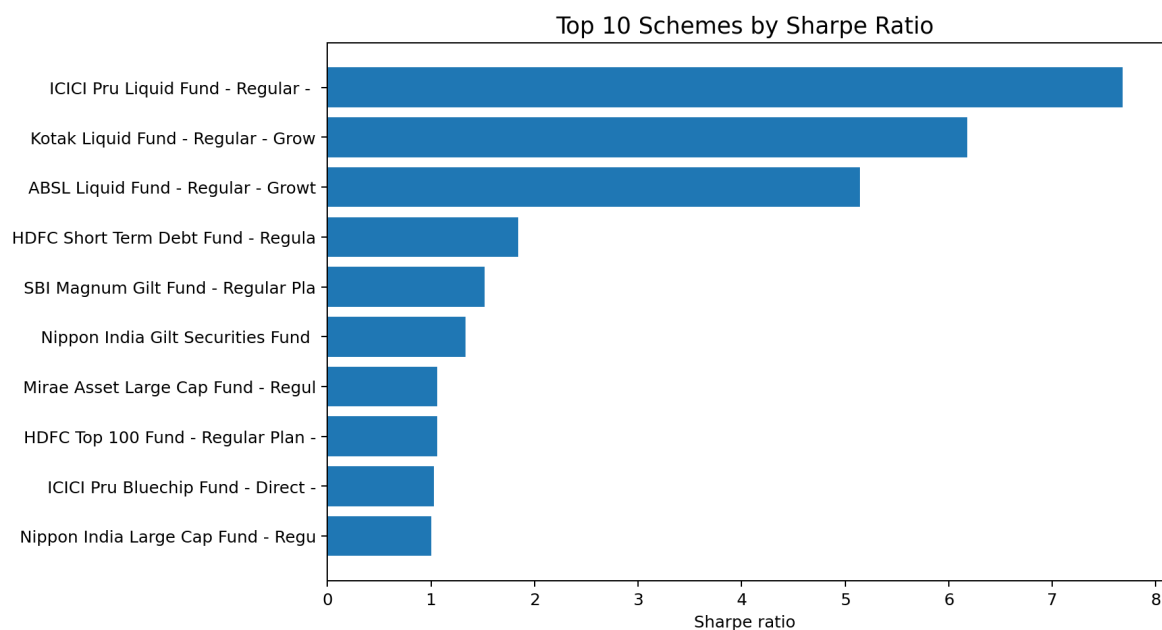


Figure 4. Top 10 schemes by Sharpe ratio in the supplied scheme-performance dataset.

The highest Sharpe ratios are reported for ICICI Pru Liquid Fund (7.68), Kotak Liquid Fund (6.18), ABSL Liquid Fund (5.14), HDFC Short Term Debt Fund (1.84) and SBI Magnum Gilt Fund (1.52). These results illustrate that Sharpe rankings can be very different from 3-year return rankings because the denominator rewards stable return paths.

The project's custom Sharpe calculation uses a 6.5% annual risk-free proxy converted to a daily rate and annualized with $\sqrt{252}$. Sortino uses only negative-return observations in the downside deviation denominator. These choices follow the specified task definitions.

9. Alpha, Beta and Maximum Drawdown

Alpha/Beta are estimated through OLS regression of fund daily returns against the supplied NIFTY100 benchmark return series using `scipy.stats.linregress`. The resulting report contains 29 schemes with sufficient overlapping observations for regression.

Scheme	Alpha	Beta	R ²	Obs.
quant Flexi Cap Fund - Growth Option-Direc	0.2765	0.0044	0.000799	295
quant Flexi Cap Fund - IDCW Option - Direc	0.2758	0.0044	0.000789	295
quant Mid Cap Fund - Growth Option - Direc	0.2451	-0.0013	0.000062	295
Nippon India Large Cap Fund - Direct Plan	0.2406	-0.0006	0.000024	295
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The largest reported annualized alpha values in the regression output are associated with quant Flexi Cap and quant Mid Cap schemes. However, the R² values are extremely low, meaning the NIFTY100 benchmark explains very little of the day-to-day return variation in these regressions. Alpha should therefore be interpreted alongside model fit, not in isolation.

Maximum drawdown identifies the largest peak-to-trough decline in each NAV path. The supplied scheme-performance reference reports the deepest drawdowns for HDFC Top 100 (-33.50%), Mirae Asset Emerging Bluechip (-33.15%), Axis Midcap (-32.38%), HDFC Mid-Cap Opportunities (-32.22%) and Nippon India Small Cap (-30.87%).

10. Advanced Risk Analytics — Historical VaR and CVaR

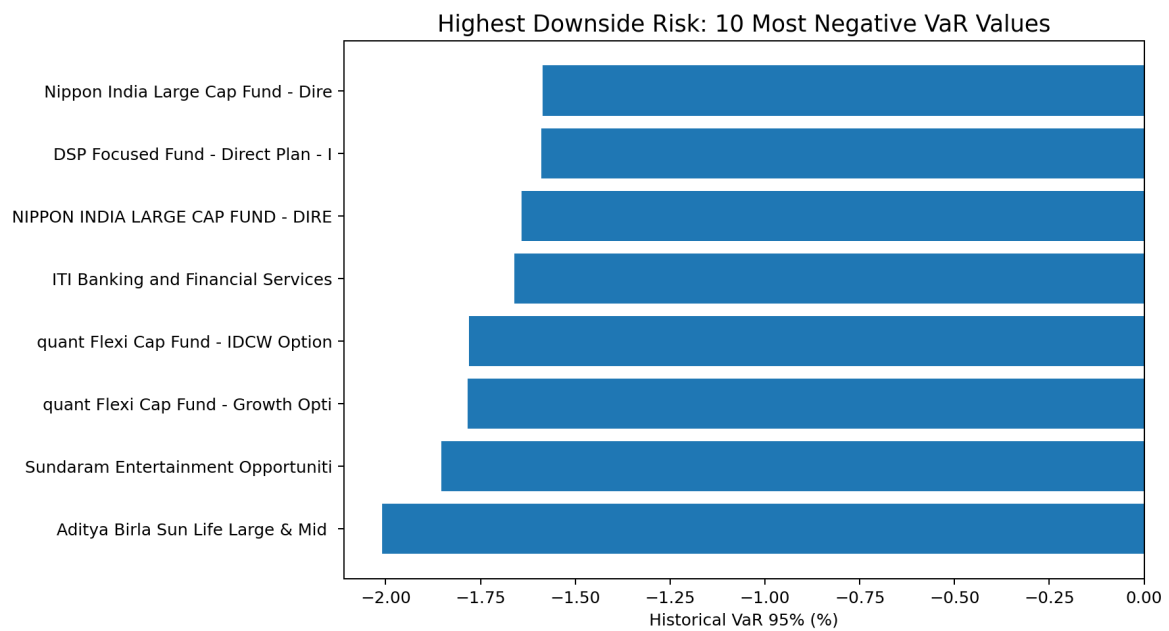


Figure 5. Most negative 95% Historical VaR values.

Historical VaR is defined as the 5th percentile of the daily-return distribution, while CVaR is the average of observations below the VaR threshold. The most negative VaR in the report is -2.01% for scheme code 100033, with CVaR of -3.21%. The next largest downside observations are -1.85% for scheme 119599 and approximately -1.78% for the quant Flexi Cap variants.

These are one-day historical loss thresholds, not forecasts. A 95% VaR of -2.01% means that approximately 5% of historical daily observations were at or below that level. CVaR provides the more severe tail-loss average and is therefore useful for comparing tail behaviour.

11. Rolling Sharpe and Investor Cohorts

The advanced notebook calculates a 90-day rolling Sharpe ratio for all 34 schemes. The rolling metric is intended to show how risk-adjusted performance changes through time rather than reducing a fund to a single full-history statistic.

The supplied notebook records 93,338 valid rolling observations. A 90-day window is used to capture medium-term changes in return and volatility while preserving enough observations for a stable time series.

Cohort analysis identifies first-transaction year and combines investor count, average SIP amount and total invested amount. The 2024 cohort is the dominant cohort by investor count in the supplied analysis. The top-fund-by-cohort output identifies scheme 119599 for 2024 and scheme 149323 for 2025 in the notebook result.

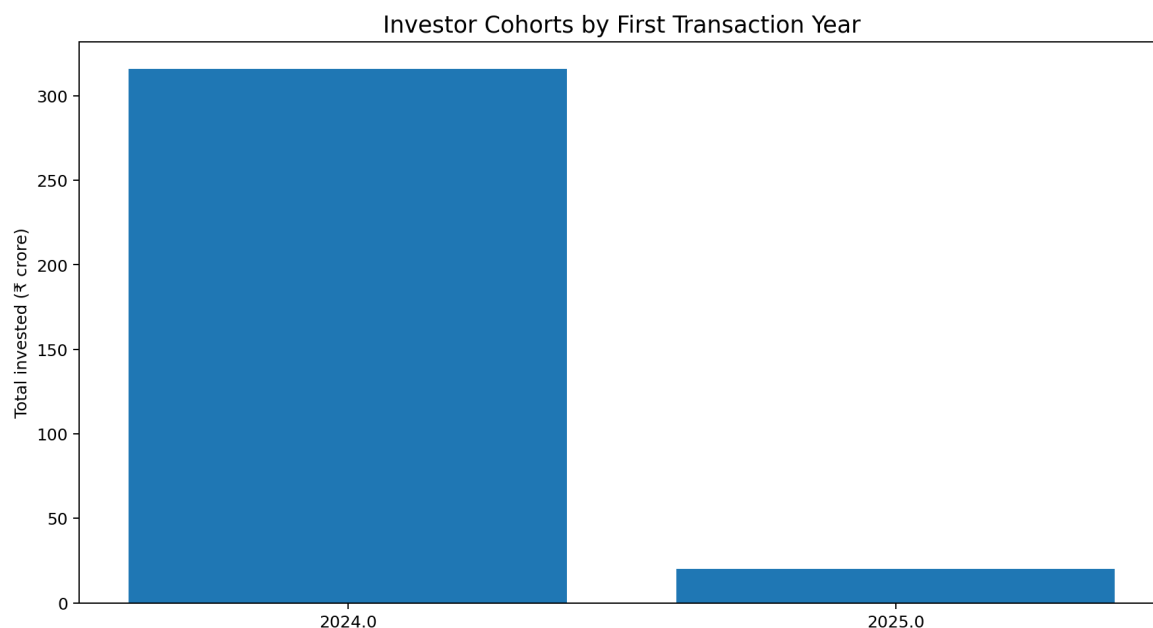


Figure 6. Cohort investment scale.

12. SIP Continuity and Sector HHI

SIP continuity is evaluated only for investors with six or more SIP transactions. The notebook identifies 38 eligible investors. Under the project rule, an average gap greater than 35 days is classified as At-Risk; all 38 eligible investors are classified At-Risk in the supplied output.

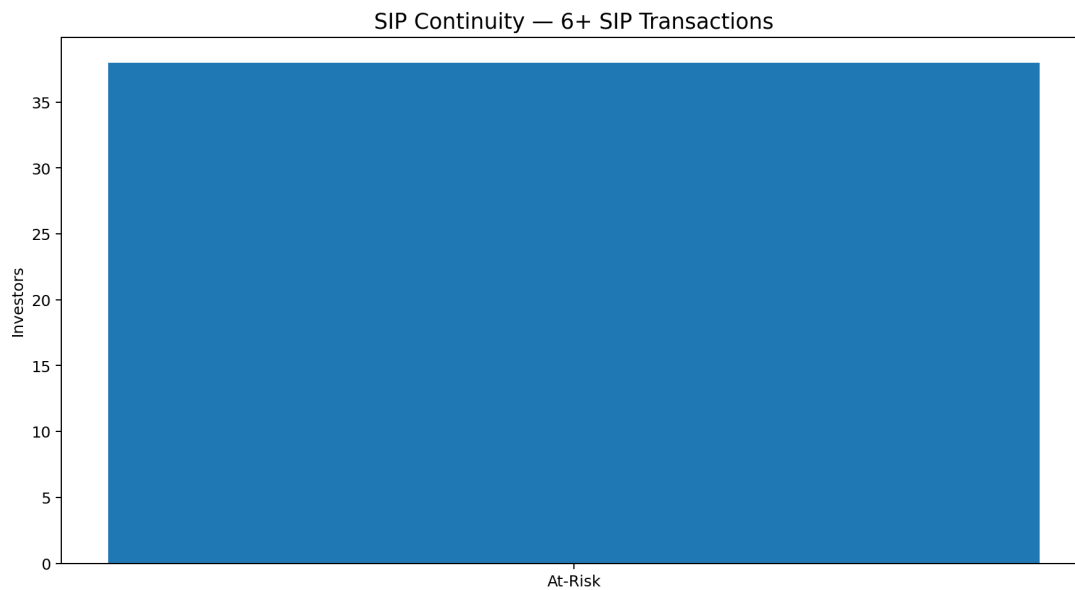
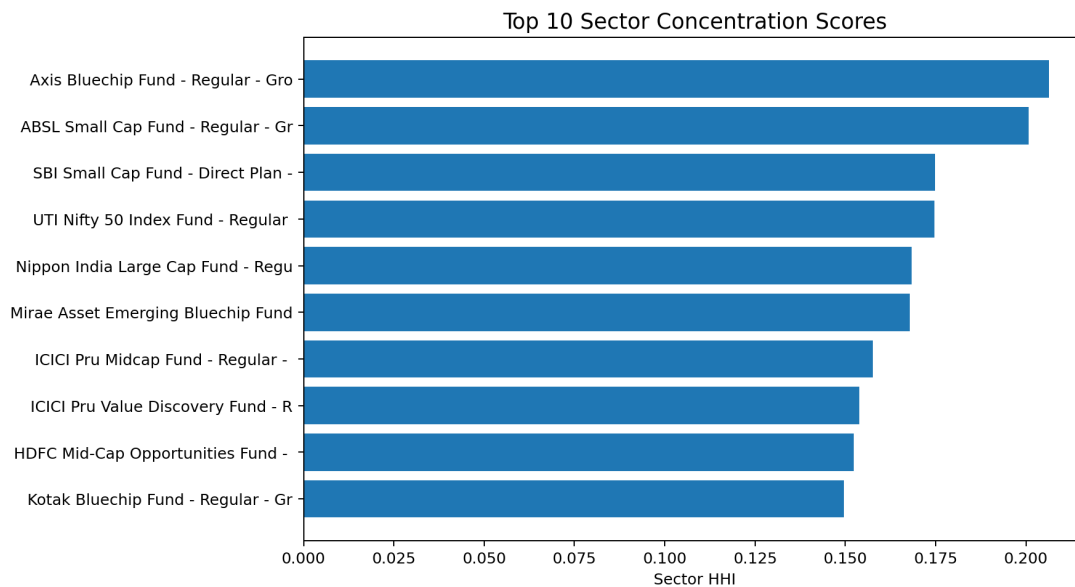


Figure 7. SIP continuity status among eligible investors.

Sector HHI measures concentration as the sum of squared sector weights. Higher values imply that a fund's portfolio is more dependent on a smaller number of sectors. The highest HHI in the report is 0.2064 for Axis Bluechip Fund, followed by ABSL Small Cap Fund at 0.2007 and SBI Small Cap Fund at 0.1748.



13. Recommender Logic and Power BI Dashboard

The recommender accepts Low, Moderate or High risk appetite and ranks matching funds by Sharpe ratio, returning up to three schemes with supporting return, Sortino, drawdown and expense-ratio fields. The final notebook snapshot exposed missing risk-grade values in one intermediate merge; the standalone recommender is designed to consume the scheme-performance table directly, where the supplied risk-grade field is populated.

The Power BI file contains four pages:

Page	Visual inventory	Purpose
Executive Overview	11	Executive KPIs, fund/category mix and trend context
Fund Performance	6	Performance trend, ranking table and fund-level comparison
Investor Analysis	7	Investor demographics, transactions and behaviour
Portfolio & Market	5	Holdings, sector concentration and market context

The PBIX layout was inspected directly from the submitted file. It contains 11 visuals on Executive Overview, 6 on Fund Performance, 7 on Investor Analysis and 5 on Portfolio & Market, with slicer visuals present on the dashboard pages. Because Power BI Desktop is not available in this execution environment, this report documents the actual PBIX page/visual structure rather than fabricating screenshots.

14. Advanced Business Insights

Insight 1 — Return leaders carry higher risk. The strongest 3-year return schemes are small-cap funds and are classified Very High risk in the supplied performance table. High return should therefore not be treated as a standalone recommendation criterion.

Insight 2 — Risk-adjusted leaders differ materially. Liquid and short-duration debt schemes dominate the Sharpe ranking, showing that stable returns can outperform volatile equity schemes on a risk-adjusted basis even when absolute returns are lower.

Insight 3 — Tail risk is measurable and fund-specific. The most negative VaR is approximately -2.01% and the corresponding CVaR is approximately -3.21%, demonstrating that average-return statistics can understate adverse-tail behaviour.

Insight 4 — SIP retention deserves attention. Among the 38 investors meeting the 6+ SIP threshold, 100% are classified At-Risk under the 35-day average-gap rule. This is an unusually strong signal for a small eligible cohort and warrants targeted retention analysis.

Insight 5 — Sector concentration varies by fund. Axis Bluechip and ABSL Small Cap have the highest sector HHI values in the supplied report. Concentration should be considered alongside performance because two funds with similar returns can have materially different diversification structures.

15. Limitations, Recommendations and Conclusion

Limitations. The supplied project contains multiple NAV snapshots: the long-form notebook file and the shorter canonical ETL output. This report explicitly distinguishes them. Benchmark coverage is also limited to the dates supplied in the benchmark dataset, so Alpha/Beta regression covers 29 schemes rather than all 40 reference performance rows. The SIP continuity result is based on a small 6+ SIP cohort. Finally, some source fields contain synthetic/reference-style values and should not be interpreted as live investment advice.

Recommendations. Standardize the canonical processed NAV snapshot used by every notebook; keep benchmark and NAV calendars aligned before regression; retain both VaR and CVaR in risk dashboards; add a cohort retention workflow for at-risk SIP investors; expose HHI and drawdown together with return metrics; and maintain a single master data dictionary so AMFI/scheme-code naming remains consistent across datasets.

Conclusion. The capstone now forms a coherent analytics product rather than a collection of isolated notebooks. The ETL pipeline executes end-to-end, SQLite provides a reproducible storage layer, EDA establishes the business context, performance analytics quantify return and risk, advanced analytics surface tail risk and investor behaviour, and Power BI provides an executive-facing interface.

Final submission package: Final_Report.pdf, Bluestock_MF_Presentation.pptx, cleaned README/repository structure and local v1.0 release tag.